

Exercise 9

Chapter 2, Page 70

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Introduction to Electrodynamics

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Solution Verified

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Step 1

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a)

The charge density is proportional to the divergence of the electric field:

$$\rho = \epsilon_0 \nabla \cdot \vec{E} = \epsilon_0 \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 K r^3) = \boxed{\epsilon_0 5r^2 K}$$

b)

Let us first find all the charge by direct integration:

$$Q = \int_V \rho dV = \int_0^R \int_0^{2\pi} \int_0^\pi \epsilon_0 5K r^2 r^2 dr \sin \theta d\theta d\phi$$

$$= 5 \frac{K \epsilon_0}{5} R^5 4\pi = \boxed{4\pi R^5 \epsilon_0 K}$$

Now let us do this via the Gauss' theorem, integrate the electric field over the sphere of radius R :

$$Q = \epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = \int_0^\pi \int_0^{2\pi} R^3 K \hat{r} \cdot R^2 \sin \theta d\theta d\phi \hat{r}$$

$$= \epsilon_0 K R^5 4\pi = \boxed{4\pi R^5 \epsilon_0 K}$$

, and both of the results agree, as they should.

Result

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a) $\boxed{\rho = \epsilon_0 5r^2 K}$

b) $\boxed{Q = 4\pi R^5 \epsilon_0 K}$

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